

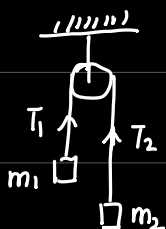
1. 滑轮的质量是 $2m$, m_1 的质量是 m , m_2 的质量是 $2m$. 求:

(1) 滑轮与绳无摩擦时, 物体的加速度;

(2) 求绳与滑轮无相对运动时的滑轮与绳子的摩擦系数临界值 μ_0 ;

(3) 当绳与滑轮摩擦系数为 $\frac{\mu_0}{2}$ 时, 求物体加速度.

解:



(1) 因为无摩擦, 则滑轮不会转动 $T_1 = T_2$

$$T_1 - m_1 g = m_1 a$$

$$m_2 g - T_2 = m_2 a$$

解: $a = \frac{1}{3}g$

(2) 有摩擦不相对滑动, 则有

$$\begin{cases} m_2 g - T_2 = m_2 a \\ T_1 - m_1 g = m_1 a \end{cases} \Rightarrow a = \frac{1}{4}g$$

$$(T_2 - T_1)R = \frac{1}{2} \cdot 2mR^2 \beta \quad (a = R\beta)$$

防止打滑: $\mu 2mgR > \frac{1}{2} \cdot 2mR^2 \beta = mR \frac{1}{4}g \Rightarrow \mu > \frac{1}{8}$, 则 $\mu_0 = \frac{1}{8}$

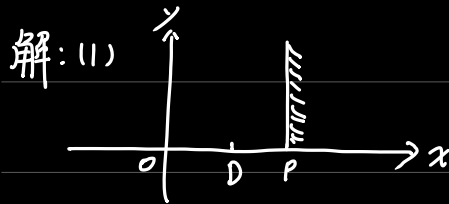
(3) $m_2 g - T_2 = m_2 a$

$$\begin{cases} T_1 - m_1 g = m_1 a \end{cases} \Rightarrow a = \frac{7}{24}g$$

$$T_2 - T_1 = \frac{\mu_0}{2} \cdot 2mg$$

2. O 点发出入射波方程为 $y = A \cos \left[2\pi \left(vt - \frac{x}{\lambda} \right) \right]$, 在 P 点发生反射, P 点后面是波密介质, 即反射点是节点. $OP = \frac{3}{4}\lambda$, $DP = \frac{1}{6}\lambda$. 求:

- (1) P 点反射波方程;
- (2) 原题丢失;
- (3) D 点驻波合成方程.



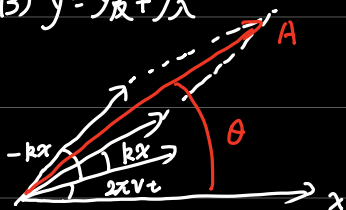
从 O 发出入射波到 P 为: $y_P^{\lambda} = A \cos \left[2\pi \left(vt - \frac{3}{4} \right) \right]$

$$= A \cos \left[\omega t - \frac{3}{2}\pi \right]$$

因为“半波损失”, 则从 P 处开始反射的波方程为:

$$y_R = A \cos \left[2\pi \left(vt + \frac{x}{\lambda} \right) \right]$$

(3) $y = y_R + y_{\lambda}$



$$y = 2A \sin \frac{2\pi}{\lambda} x \cos 2\pi vt$$

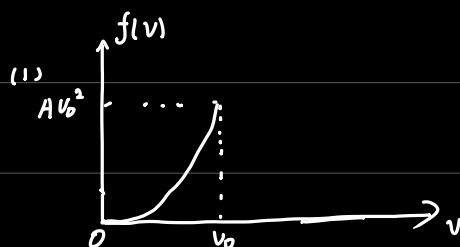
D 处的驻波方程为: $y_D = \sqrt{3}A \cos 2\pi vt$

3. 电子气由 N 个自由电子组成, 电子速率在 $v \sim v + dv$ 之间的概率

为 $\frac{dN}{N} = \begin{cases} Av^2 dv, & 0 \leq v \leq v_0 \\ 0, & v > v_0 \end{cases}$, 式中 A 为常量.

- (1) 做出速率分布曲线;
- (2) 用 v_0 定出 A ;
- (3) 求 $v_p, \bar{v}, \sqrt{\overline{v^2}}$;
- (4) 求速率分布在 0 到 $\frac{v_0}{2}$ 间的电子方均根速率.

解: $f(v) = \begin{cases} Av^2, & 0 \leq v \leq v_0 \\ 0, & v > v_0 \end{cases}$



$$(2) 1 = \int_0^{v_0} A v^2 dv \Rightarrow A = \frac{3}{v_0^3}$$

$$(3) v_p = v_0$$

$$\bar{v} = \int_0^{v_0} v A v^2 dv = \frac{3}{4} v_0$$

$$\sqrt{v^2} = \sqrt{\int_0^{v_0} v^2 A v^2 dv} = \sqrt{\frac{3}{5}} v_0$$

$$(4) \sqrt{v^2} = \left(\int_0^{v_0} A v^4 dv \right)^{\frac{1}{2}} = \frac{1}{4} \sqrt{\frac{3}{10}} v_0$$

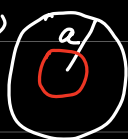
4. 一圆柱高 l , 半径为 a , $l \gg a$. 无限长. 体电荷密度为 ρ , 在一外力下, 匀加速转动, 即 $\omega = \beta t$, 求

(1) 圆柱内电流;

(2) 圆柱内 $B(t)$;

(3) 圆柱内 $E(t)$.

解: (1)



$$dV = 2\pi r dr l, \quad dq = \rho dV$$

$$\Rightarrow I(r, t) = \int_0^r \frac{dq}{2\pi r} = \frac{1}{2} \rho l r^2 \beta t$$

$$(2) 2\pi r \cdot B(r, t) = \mu_0 I(r, t) \Rightarrow B(r, t) = \frac{\mu_0 l r \rho \beta t}{4\pi}$$

$$(3) 2\pi r \cdot E(r, t) = -\frac{d}{dt} B(r, t) \Rightarrow E(r, t) = \frac{\mu_0 l r \rho \beta}{8\pi^2}$$

5. 以每毫米有 500 条栅纹的衍射光线观察钠光谱线 ($\lambda = 590 \text{ nm}$), 缝宽 a 和刻痕 b 相等, 试问:

(1) 平行光垂直入射于光栅时能看到第几级光谱线?

(2) 平行光以与光栅平面法线夹角为 30° 入射时, 能看到几级光谱线?

解: (1) $d = \frac{1 \text{ mm}}{500 \text{ 条}} = 2000 \text{ nm} \Rightarrow \frac{d}{a} = 2$

$a = 1000 \text{ nm}$

(2) 略, 原题

(3) 略.

6. 小球半径为 a , 大球半径为 b 的同心球, 小球带电 Q , 电导率为 σ , 求:

(1) $Q(t)$ 表达式;

(2) 求产生的焦耳热并证明其等于静电做的功.

解: (1) $a < r < b$ 间电场为 $E = \frac{Q(t)}{4\pi\epsilon_0 r^2}$, 由微分形式的欧姆定律有:

$$\text{电流密度 } j = \sigma E = \frac{\sigma}{4\pi\epsilon_0} \frac{Q(t)}{r^2}$$

$$\text{则同心球之间的电流 } I = 4\pi r^2 j = \frac{\sigma}{\epsilon_0} Q(t) = -\frac{d}{dt} Q(t)$$

$$\text{解得: } Q(t) = Q e^{-\frac{\sigma}{\epsilon_0} t}$$

(2) 微分形式焦耳定律: 热功率密度 $w = \sigma E^2$, 则

$$\text{焦耳热为 } Q = \int_0^t \int_V w dV dt = \frac{Q^2}{8\pi\epsilon_0} (1 - e^{-\frac{2\sigma}{\epsilon_0} t}) \left(\frac{1}{a} - \frac{1}{b}\right)$$

静电做功: 当电场为 $E(t)$ 时, 将电荷移到 | 大球上作功微分.

$$\text{电容 } C = \frac{4\pi\epsilon_0 ab}{b-a} \quad (\text{易求})$$

$$\text{则静电力做功: } W = \int_Q^{Q(t)} \frac{q}{C} dq = \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b}\right) (e^{-\frac{2\sigma}{\epsilon_0} t} - 1)$$

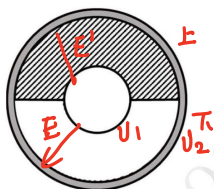
$|W| = |Q|$, 证毕.

7. 如图所示, 小球和薄球壳组成系统, 球壳内一半为电解质 ϵ_r , 小球半径 R_1 , 球壳半径 R_2 , 小球带电 Q . 求:

(1) 内球壳电量分布;

(2) 电势差 $U_1 - U_2$;

(3) 求 U_1 .



解: 假设上半球、下半球的电荷均匀分布, 则有

$$\begin{cases} E = \frac{Q_{\text{下}}}{4\pi\epsilon_0 R^2} \\ E' = \frac{Q_{\text{上}}}{4\pi\epsilon_0 \epsilon_r R^2} \end{cases}$$

(1) 因为 $U_1 - U_2 = \int_{R_1}^{R_2} E \, dR = \int_{R_1}^{R_2} E' \, dR$, 则有 $E = E'$

$$\begin{cases} Q_{\text{下}} = \frac{Q_{\text{上}}}{\epsilon_r} \\ Q_{\text{下}} + Q_{\text{上}} = Q \end{cases} \Rightarrow \begin{cases} Q_{\text{上}} = \frac{\epsilon_r Q}{1 + \epsilon_r} \\ Q_{\text{下}} = \frac{Q}{1 + \epsilon_r} \end{cases}$$

$$\begin{aligned} (2) \quad U_1 - U_2 &= \frac{Q_{\text{下}}}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ &= \frac{Q}{4\pi\epsilon_0 (1 + \epsilon_r)} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \end{aligned}$$

$$(3) \quad U_1 = \int_{R_1}^{R_2} E \, dR + \int_{R_2}^{\infty} \frac{Q}{4\pi\epsilon_0 R^2} \, dR = \frac{Q}{4\pi\epsilon_0 (1 + \epsilon_r)} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) + \frac{Q}{4\pi\epsilon_0 R_2}$$